WEEK 10 — TESTING, DEBUGGING & QUALITY ASSURANCE

Test Cases

Course	CSE 4104 — Software Development III
Team	CSE4104-7C-T05
Project Title	AI Academic Mentor
Instructor	Md. Riaz Mahmud, Assistant Professor, Dept. of CSE, NUBTK
Section	7C
Submission Date	07 September 2026
GitHub Repository Link	GitHub - Priyo-1158/cse4104-7c-t05-ai-academic-mentor: AI Academic Mentor - Personalized Learning Assistant for Students · GitHub

Team Information

Role	Member Name	Student ID	Contribution
Team Leader	Shadman Sadequeen Priyo	11230121158	Project coordination, GitHub management, integration testing
Frontend Developer	Musabbir Hossain Chayon	11230121168	React/Vite UI, API wiring, report authoring, deployment
Backend Developer	Sk. Nadirul Haque Nadir	11230121155	Express routes, MongoDB models, JWT authentication
AI Integration Lead	Shafin Kabir	11230121175	Gemini/HuggingFace routes, prompt design

ID	Feature	Scenario	Input	Status
TC-01	Registration	Register with valid data	name, valid email, 8-char password	**PASS**
TC-02	Registration	Register duplicate email	Same email as TC-01	**PASS**
TC-03	Registration	Register with missing fields	{ email only }	**PASS**
TC-04	Registration	Register with short password	3-char password	**PASS**
TC-05	Registration	Register with invalid email format	"not-an-email"	**PASS**
TC-06	Registration	Request role=admin without access code	role: "admin", no code	**PASS**
TC-07	Registration	Request role=admin with correct code	role: "admin" + code from .env	**PASS**
TC-08	Registration	Register with empty body	{}	**PASS**
TC-09	Login	Login with valid credentials	Correct email + password	**PASS**
TC-10	Login	Login with wrong password	Correct email, wrong password	**PASS**

ID	Feature	Scenario	Input	Status
TC-11	Login	Login with non-existent email	Unregistered email	**PASS**
TC-12	Login	Login with empty fields	email: "", password: ""	**PASS**
TC-13	Auth / Session	GET /auth/me with a valid token	Valid Bearer token	**PASS**
TC-14	Auth / Session	GET /auth/me with no token	(no Authorization header)	**PASS**
TC-15	Auth / Session	GET /auth/me with a garbage token	Bearer garbage.token.value	**PASS**
TC-16	Auth / Session	Malformed Authorization header	Authorization: <raw token>	**PASS**
TC-17	Authorization / RBAC	Student attempts /admin/stats	Valid student-role token	**PASS**
TC-18	Authorization / RBAC	Admin accesses /admin/stats, DB unavailable	Valid admin-role token	**PASS**
TC-19	Authorization / RBAC	Student attempts /admin/users list	Valid student-role token	**PASS**
TC-20	Logout	End a user session	N/A — client-side action	**PASS**
AI-01	AI Chat	Normal input	Well-formed question	**PASS**

ID	Feature	Scenario	Input	Status
AI-02	AI Chat	Very short input	"OS?"	**PASS**
AI-03	AI Chat	Very long input (~600 words)	~4,300-character prompt	**PASS**
AI-04	AI Chat	Invalid input (empty messages array)	{ messages: [] }	**PASS**
AI-05	AI Chat	Ambiguous input	"it"	**PASS**
AI-06	AI Chat	Unexpected input (digits only)	"12345 67890"	**PASS**
AI-07	AI Chat	Repeated input (1st send)	"What is DBMS?"	**PASS**
AI-08	AI Chat	Repeated input (2nd, identical send)	"What is DBMS?"	**PASS**
AI-09	AI Chat	Empty input	content: ""	**PASS**
AI-10	AI Quiz	Domain-specific: generate a 3-question quiz	topic: "Operating Systems", numQuestions: 3	**PASS**
AI-11	AI Study Plan	Domain-specific: generate a 3-day plan	subjects: [DBMS, OS], days: 3	**PASS**

ID	Feature	Scenario	Input	Status
AI-12	AI Chat	Adversarial / edge case: script-injection payload	<script>alert(1)</script> && DROP TABLE users;	PASS
TC-27	Quiz CRUD	GET /api/quiz without auth	(no token)	PASS
TC-28	Quiz CRUD	POST /api/quiz missing required fields	topic only	PASS
TC-29	Quiz CRUD	POST /api/quiz with valid data	topic, score, numQuestions, correct	PASS
TC-30	Quiz CRUD	GET /api/quiz after saving a result	Valid token	PASS
TC-31	Notes CRUD	GET /api/notes/:id — non-existent ID	A well-formed but unused ObjectId	PASS
TC-32	Notes CRUD	GET /api/notes/:id — malformed ID	Non-ObjectId string	PASS
TC-33	Routing	Request an undefined API route	/api/totally/not/a/real/route	PASS
TC-34	Authorization / RBAC	Admin sets a role via PUT, DB unavailable	{ role: "superadmin" }	PASS
TC-35	Security / CORS	Preflight OPTIONS from a disallowed origin	Origin: https://evil-site.com	BLOCKED

ID	Feature	Scenario	Input	Status
TC-36	Security / CORS	GET request from a disallowed origin	Origin: https://evil-site.com	BLOCKED
TC-37	UI/UX — Navigation	Click through primary navigation	N/A — requires browser	TESTED
TC-38	UI/UX — Forms	Submit forms with valid/invalid data via UI	N/A — requires browser	TESTED
TC-39	Responsive — Mobile	View Login page at mobile width	N/A — requires device/emulator	PASS
TC-40	Data deletion	Delete a note/quiz/plan record	N/A	NOT APPLICABLE
TC-41	Search / Filter	Search or filter records by keyword	N/A	NOT APPLICABLE